A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived causal structure

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Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce χ-VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A mechanical audit finds no hidden incompatible operation, and local reconstructions agree near the float32 precision floor. On all ten LIBERO-Object tasks, the 20.1M-parameter policy reaches $92.8\% \pm 2.1\%$ with a ViT encoder and $93.7\% \pm 0.8\%$ with a convolutional encoder under the published three-seed protocol. A same-skeleton conventional twin differs by only 0.006% in parameters. Complete same-hardware replications find a 4.4-point rational deficit on A30 and 5.8 points on A6000. Compilation reduces the latency ratio from 3.49–$5.33\times$ in eager mode to 1.73–$1.74\times$ across the two GPU classes. We derive an exact weight-based decomposition of softmax-free attention and reconstruct every attention module in three checkpoints with worst relative error 2.56×10^{-7}. All 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. At the policy's visual bond, retaining 64 of 384 causally ranked dimensions succeeds in $29/60$ matched trials across three checkpoints, compared with $47/60$ for the full representation and $0/60$ for random subspaces. The causal effect replicates, but near-lossless preservation does not. A 300-pair instruction intervention across three checkpoints shows consistent language-driven motion but incomplete scene-prior override. The evidence establishes competitive specialist capability and exact layerwise access while exposing nonzero seed-0 capability and systems costs. It does not establish broad generalization or selective coefficient-level repair.

1 Introduction

Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains millions of learned numbers, but it does not directly reveal which combinations of image regions, instruction tokens, and robot state produce an action. Analysts therefore usually collect activations on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the same structure is present in the weights and whether it causes behavior.

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Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. Decomposing that table can rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic form. The aim is not merely compression. It is a direct path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

> *Can a specialist VLA retain competitive closed-loop capability when every deployed learned operation is polynomial or rational, and does the resulting structure provide a useful weight-based analysis surface?*

The stronger thesis has three parts: decomposability should have small capability cost, the weights should expose stable behaviorally causal interactions, and editing those interactions should selectively change behavior without retraining. This paper establishes strong capability and exact layerwise structure, then states clearly where causal diagnosis and editing remain incomplete.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%. A same-skeleton conventional twin differs by only 0.006% in parameters. Two complete same-hardware replications place the seed-0 capability cost between 4.4 and 5.8 points.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. We show a causal closed-loop consequence. A selected 64-dimensional visual subspace consistently outperforms random equal-width subspaces across three checkpoints, while remaining materially below the full policy. A separate three-checkpoint instruction intervention distinguishes reliable trajectory response from incomplete task-level rerouting.

Table 1 separates the evidence already established from the tests still needed for the stronger thesis. This distinction is important because algebraic representability does not by itself imply tractable analysis, and a causally important activation subspace is not yet a coefficient-level editing interface.

Conjunct	Current evidence	Decisive missing test
Small capability cost	The rational policy reaches $92.8\% \pm 2.1\%$ over three seeds. A matched seed-0 twin shows a 4.4 to 5.8 point cost across two GPU classes.	Diagnose the collapsed conventional seeds 1 and 2, evaluate matched χ-VLA seeds, and add partial-conversion controls.
Weight-only diagnosis	Exact layerwise attention decomposition consistently beats fixed random subspaces.	Recover a concrete behavioral condition from weights before viewing its test rollouts.
Predictive surgery	A selected activation subspace causally preserves behavior.	Edit specified coefficients, predict the direction, and measure target effect and collateral damage without retraining.

Table 1: **Status of the three-part thesis.** Strong capability and exact layerwise structure are established. Hardware-controlled cost and coefficient surgery remain open.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho}L_{\rho i}R_{\rho j}\right)\bar{x}_i\bar{x}_j. \tag{1}$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1K_1^\top) \odot (Q_2K_2^\top)/s\right]V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike softmax attention, every multiplicative interaction can be expanded into coefficients determined by the trained weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z+\epsilon}$:

$$\mathrm{RNorm}(x) = x\, r\left(\frac{1}{d}\sum_{j=1}^{d} x_j^2\right). \tag{3}$$

Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D. Linear, bilinear, residual, and normalization operations update this pair. The model therefore keeps input-specific rescaling while remaining exactly representable as a rational tensor network.

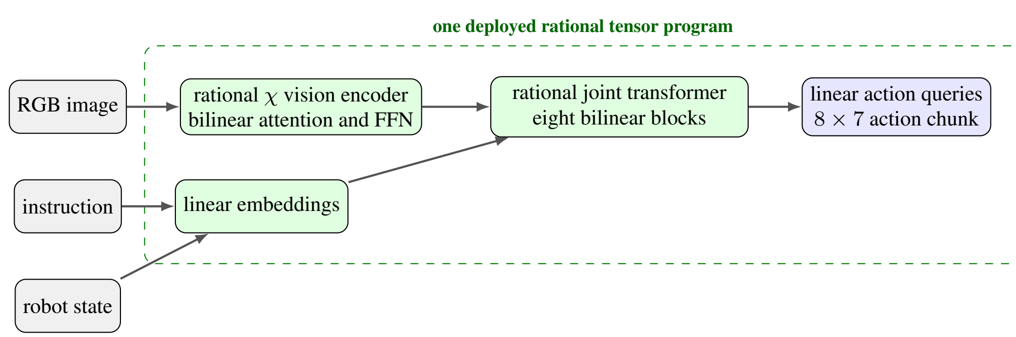

Figure 1: **Deployed χ-VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has $20{,}137{,}352$ learned parameters.

Training uses $66{,}984$ demonstration frames, $40{,}000$ AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real $189/200$ checkpoint at runtime and reconstruct representative operations from saved weights.

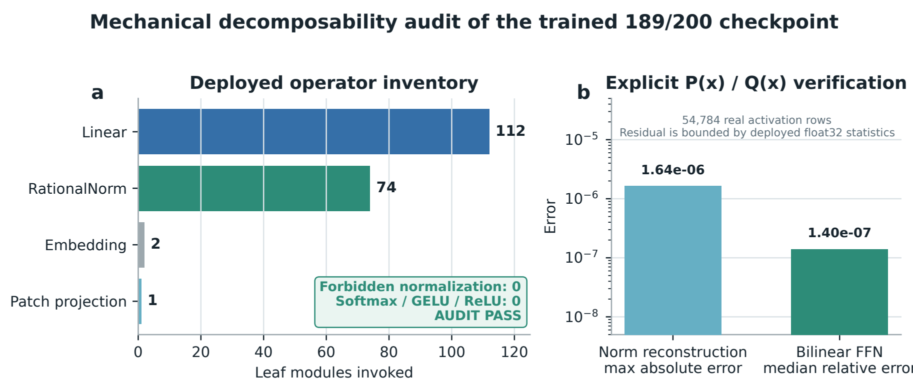

Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over $54{,}784$ real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and $1{,}500$ rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

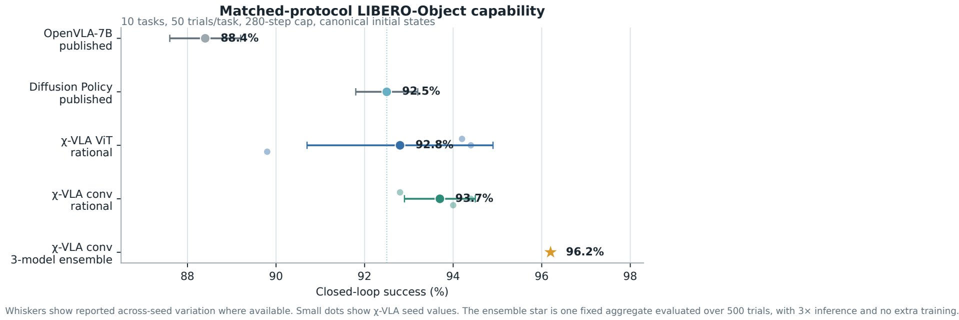

Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are χ-VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ-VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
Same-skeleton conventional ViT	20.1M	1	89.8%
χ-VLA, rational conv	20.1M	3	$\mathbf{93.7\% \pm 0.8\%}$
χ-VLA, conv ensemble	3×20.1M	$3 \to 1$	$\mathbf{96.2\%}$
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 2: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

The same-skeleton conventional control replaces bilinear attention, bilinear FFNs, and rational normalization with softmax attention, SwiGLU, and RMSNorm. It keeps the observations, token layout, demonstrations, optimizer, updates, seed, action decoder, canonical initial states, and trial budget fixed. On A30, the conventional policy succeeds in $448/500$ trials and the rational policy in $426/500$. On A6000, they reach $451/500$ and $422/500$. The rational model has $20{,}137{,}352$ parameters and the conventional model has $20{,}138{,}632$, a difference of 0.006%. The rational deficit is therefore 4.4 and 5.8 points. A stratified trial bootstrap gives intervals of $[-8.0, -0.8]$ and $[-9.4, -2.2]$ points, respectively. These intervals quantify the fixed-checkpoint canonical states, not across-seed uncertainty. Task 3 accounts for 21 of the lost successes on both platforms. Targeted canonical replications reverse this gap: χ-VLAseeds 1 and 2 each reach $41/50$, while their conventional counterparts reach $37/50$ and $33/50$. The seed-0 task-3 deficit is therefore not directionally consistent across seeds, although a matched native training matrix is still pending. The full seed-1 χ-VLA checkpoint reaches $404/500$ trials, or 80.8%, while the seed-1 conventional checkpoint reaches $45/500$. Their different trainer provenance prevents interpreting this as conversion cost. The two collapsed conventional checkpoints nevertheless attain training-cache normalized MSE of 9.51×10^{-4} and 1.19×10^{-3}. This rules out gross checkpoint corruption, but not closed-loop covariate shift or held-out generalization failure. A portability control places the same χ-VLA checkpoint at $423/500$, or 84.6%, on A40, between its A30 and A6000 results. The A40 conventional matrix is still running. On one A30, eager batch-one latency is 38.05 ms for the rational model and 7.14 ms for the conventional control. Compilation reduces these to 1.52 and 0.88 ms, a $1.74\times$ gap. At batch 128, rational training is $3.12\times$ slower and uses $1.63\times$ the peak allocated memory. A6000 profiling independently reduces the eager $3.49\times$ gap to $1.73\times$ after compilation.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability and closes the first same-seed capability-cost comparison. One conventional seed is not enough for a three-seed noninferiority claim, and the eager latency gap is material. Native matched seeds, partial-conversion controls, and optimized execution are still required before claiming that convertibility is broadly or computationally free.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient table. Each entry says how strongly one combination of query, key, and value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5}. This residual comes from the deployed implementation's intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force materialization to 2.13×10^{-14}.

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer's computation.

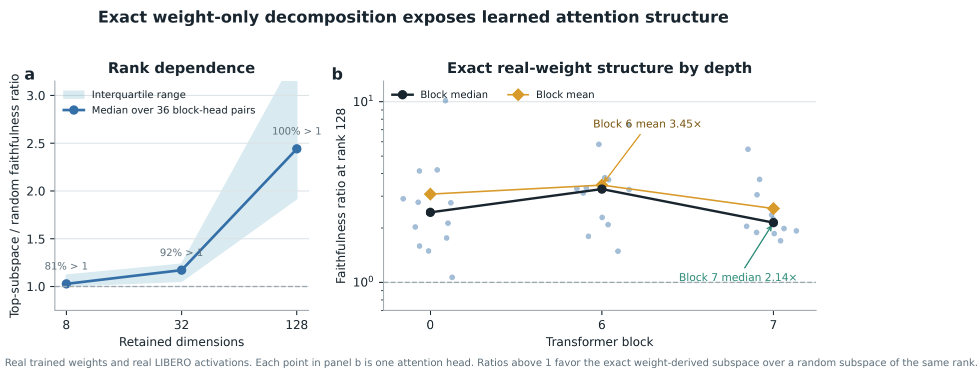

Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure and a language shortcut

We next test whether the identified structure affects closed-loop behavior and whether high task success reflects robust use of the language input.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. A subspace is a smaller coordinate system inside the model's 384-dimensional visual representation. Retaining a subspace means zeroing every direction outside it, then running the actual policy. We compare the selected directions with random equal-width controls in offline reconstruction and closed-loop intervention. This ranking uses cached training demonstrations and downstream action gradients. The offline reconstruction is therefore an in-cache diagnostic. The closed-loop intervention uses independent canonical simulator states, so it tests a causal bottleneck but is not a weight-only discovery result.

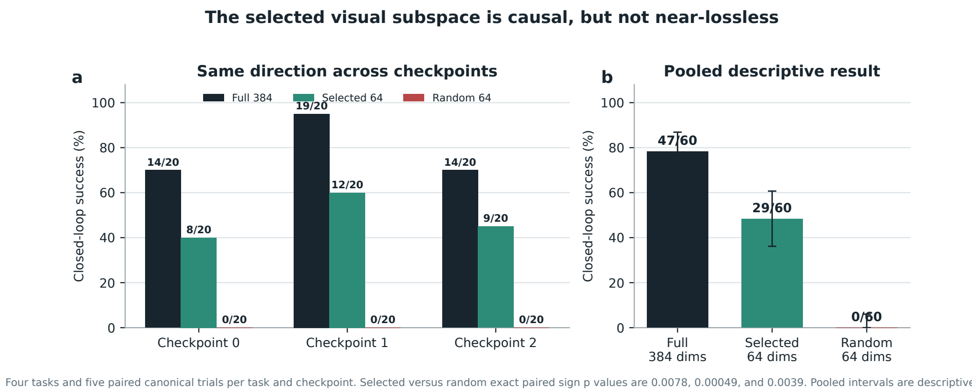

Figure 5: **The selected visual subspace has a replicated causal effect, but it is not near-lossless.** Each checkpoint uses four tasks and five paired canonical trials per task. Selected subspaces beat random controls in all three checkpoints. The pooled interval is descriptive because trials within a checkpoint share learned weights.

An earlier one-checkpoint screen found in-cache random-to-selected reconstruction ratios of $4.8\times$, $4.6\times$, and $41.8\times$ for translation, rotation, and gripper, followed by $19/20$, $18/20$, and $0/20$ closed-loop successes for full, selected, and random conditions. The three-checkpoint replication is weaker but more decisive. Full representations succeed in $14/20$, $19/20$, and $14/20$ trials. Selected subspaces reach $8/20$, $12/20$, and $9/20$, while all random controls reach $0/20$. Exact paired sign tests for selected versus random give $p = 0.0078$, 0.00049, and 0.0039. The selected substrate is therefore causally important and replicates in direction, but its $29/60$ pooled success is 30 points below the full policy's $47/60$. The earlier near-lossless result does not replicate. Offline fidelity from the new runs is not cited because the completed jobs reused discovery samples for evaluation. Disjoint-sample diagnostics are queued. The exact layerwise attention method in Section 4 must still be connected to the same full-suite closed-loop intervention.

5.2 Counterfactual instruction test

High success does not guarantee robust language use. A one-step screen is weak: replacing the instruction with one naming another visible object shifts the predicted reach direction toward that object in only 15.0% of 80 canonical states. We therefore run paired 80-step rollouts from 100 matched canonical states for each of three checkpoints. The instruction change increases relative end-effector preference for the renamed object by 0.175, 0.168, and 0.189 m. All three trial-bootstrap 95% intervals exclude zero. The trajectories shift toward the renamed object in 84%, 86%, and 88% of pairs, but end closer to it in only 33%, 28%, and 28%. Appendix Figure 7 reports the replication. The seed-0 conventional twin responds even more strongly, with a 0.239 m shift, 92% directional response, and a 33% endpoint rate. Instruction responsiveness is therefore not unique to the algebraic architecture. This is behavioral grounding evidence, not counterfactual task success, because the benchmark predicate still names the original object and the horizon covers the reach phase.

The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target, so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy range.

This mixed result narrows the VLA claim. Language changes the trajectory reproducibly, but scene priors often prevent complete rerouting. The model is therefore a capable language-conditioned specialist, not yet a robust instruction-grounded policy. LIBERO-CF independently documents the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively repair that shortcut.

6 Decisive remaining tests

1. Complete the controlled conversion study. Conventional seeds 1 and 2 completed but reached only 9.0% and 8.2%, despite low final-minibatch losses, compared with 89.8% for seed 0. Audit this severe closed-loop seed sensitivity and checkpoint provenance. The strong seed-0 checkpoint is a legacy artifact, while seeds 1 and 2 were trained by the Athena-native trainer. Evaluate a fully native matrix and the matched χ-VLA checkpoints, add partial-conversion controls, and retain the predeclared three-point noninferiority margin. Seed 0 misses that margin on both tested GPU classes. Report FLOPs, training throughput, memory, eager latency, and optimized latency separately. The current evidence supports strong rational capability and a modest nonzero seed-0 cost.

2. Direct coefficient surgery. Our preregistered discovery sweep evaluates fifteen block-rank configurations and directly edits query, key, and value columns selected by exact attention Grams. No configuration passes both 15-point gates. The best minimum advantage is block 0 at rank 128, where retention beats the matched random control by 10 points and selected removal is 40 points more damaging than matched random removal. This is a no-go for the current edit mapping, as detailed in Appendix A.7. The next test must resolve redundant paths or gauge sensitivity, compare post-hoc controls across three checkpoints, and report targeted effect, ordinary success, edit norm, and collateral damage.

3. Concrete diagnosis and selective repair. Develop two capstones. Gripper timing is the lower-risk target because phase transitions and premature closure have explicit simulator variables. Scene identity versus instruction binding is the higher-upside target because the current policy demonstrably fails the counterfactual test. In both cases, discover the algebraic term from weights, attach a limited semantic label, freeze the candidate, predict the behavioral consequence, and test on blind paired rollouts. A compelling repair should improve the targeted failure by 10 to 20 points, lose no more than 2 to 3 points of ordinary success, and preserve its direction across three seeds.

4. Broader capability and grounding. Evaluate the conventional and rational finalists on LIBERO-Goal and LIBERO-Long, followed by a counterfactual grounding benchmark such as LIBERO-CF [10]. Object-only success is insufficient because standard LIBERO can reward scene-to-action memorization [9]. Test the stability of recovered mechanisms under equivalent reparameterizations and across trained seeds.

5. RL only after a paired gate. The repaired-reward AWR pilot improves 80.7% to 82.7%, but it has one seed and fifteen trials per task. A justified follow-up needs three paired seeds, a matched-compute behavior-cloning continuation control, full protocol evaluation, and regression gates. Generic PPO on saturated LIBERO-Object is lower priority than the four tests above.

7 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ-VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ-nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight-derived discovery and direct coefficient edits in a competitive continuous vision-language policy, evaluated for stability and collateral damage against matched post-hoc baselines.

VLA robustness. Recent evaluations show that standard LIBERO success can coexist with weak robustness and language use [9–11]. Our instruction intervention reaches the same conclusion from an independently trained compact specialist and motivates a decomposition-guided diagnosis.

8 Limitations and conclusion

χ-VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis now crosses individual attention layers, and a compact causal visual subspace consistently beats random controls across three checkpoints without preserving the full policy near losslessly.

Five limits determine the next version of the paper. First, the same-skeleton seed-0 control has a 4.4 to 5.8 point capability gap and a residual compiled-runtime tax. Two additional conventional seeds collapse to 9.0% and 8.2%, exposing severe closed-loop seed sensitivity rather than establishing a stable multi-seed comparison. Their Athena-native provenance also differs from the strong legacy seed-0 checkpoint. Second, the benchmark is one narrow suite and counterfactual instructions do not fully reroute the policy. Third, exact decomposition is layerwise rather than end to end. Fourth, the three-checkpoint closed-loop intervention establishes a causal bottleneck but loses 18 of the full policy's 47 successes. Fifth, activation projection succeeds but the full fifteen-configuration coefficient-edit sweep has no candidate that passes both selectivity gates, so no behaviorally validated weight edit has yet been established.

The defensible conclusion is therefore narrow. Polynomial and rational restrictions do not prevent a compact policy from reaching competitive LIBERO-Object capability. The matched seed-0 control isolates a modest nonzero cost that now requires multi-seed explanation. The restrictions also provide a precise weight-derived analysis surface. Establishing multi-seed conversion cost, broad language-grounded control, semantic diagnosis from weights, and a decomposition-enabled repair requires the controlled experiments in Section 6. If all three conjuncts in Table 1 hold on the same trained policies, the result is stronger than any individual capability, decomposition, or intervention claim.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

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A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in $189/200$ trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

A.2 Matched architecture capability and systems cost

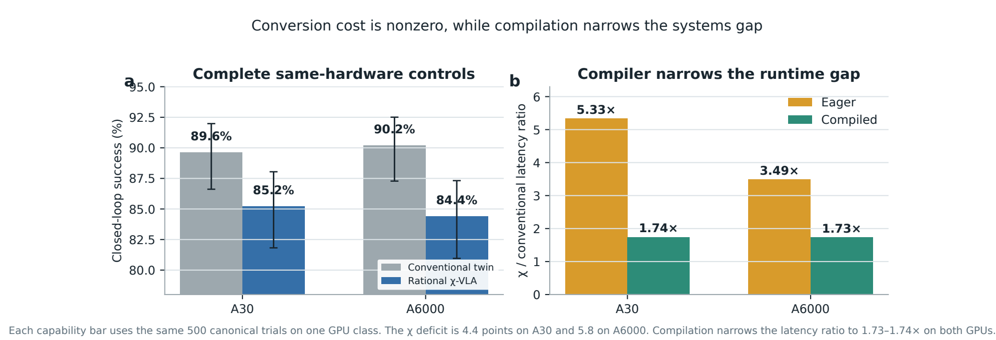

Figure 6: **Conversion cost is modest but nonzero, and compilation narrows the systems gap.** Complete same-hardware evaluations give 89.6% versus 85.2% on A30 and 90.2% versus 84.4% on A6000, conventional then rational. Parameters differ by 0.006%. Compilation narrows the batch-one latency ratio to 1.73–$1.74\times$ on both GPUs.

A.3 Three-checkpoint counterfactual grounding

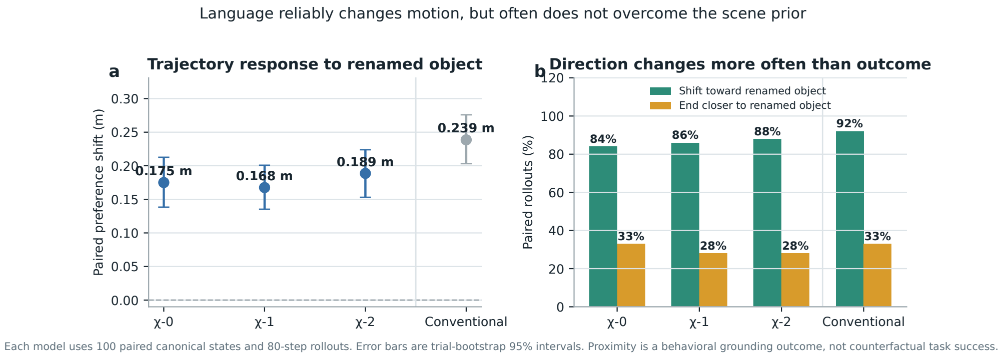

Figure 7: **Renaming the target changes motion reliably but does not fully override the scene prior.** Each model is evaluated on 100 paired canonical states for 80 control steps. The left panel reports the change in end-effector preference for the newly named object, with 95% bootstrap intervals. The right panel separates directional response from the stricter condition of ending closer to that object. These are proximity tests, not counterfactual task-success measurements. The conventional control shows that ordinary language response is not unique to tensor decomposability.

A.4 Per-task ensemble behavior

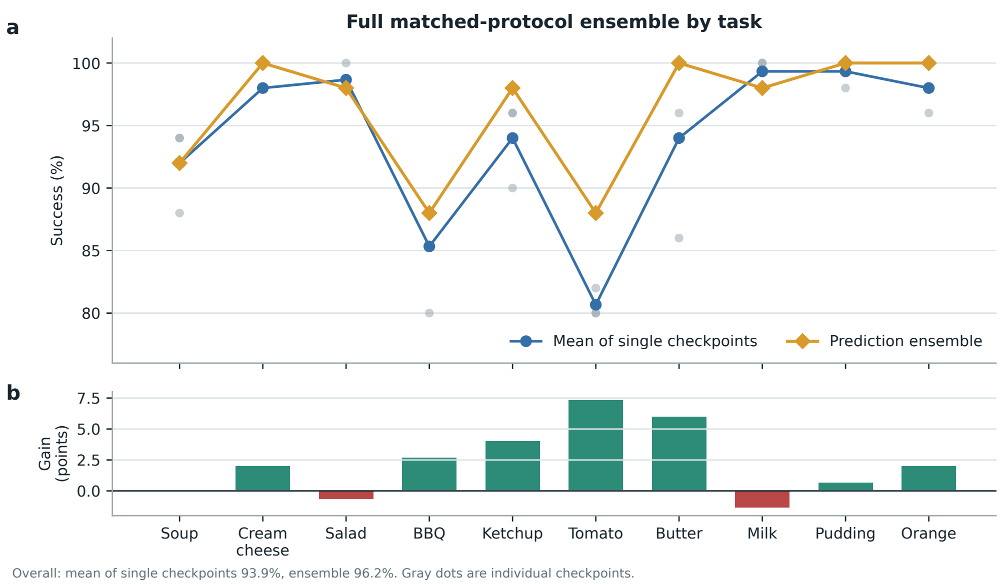

Figure 8: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

A.5 Product-routing multimodal head

For an action-query state h, the product-routing head represents

$$a(h,b) = c_0(h) + \sum_{g=1}^{G} (2b_g - 1)c_g(h), \qquad b \in \{0,1\}^G. \tag{4}$$

The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$ and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head. Product routing remains a construction result until it wins a controlled genuinely multimodal task.

A.6 Development interventions

Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven corrective episodes and $1{,}131$ frames. Applying the same update thinly across all ten tasks reduces overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points. A reward audit shows that accumulated step cost makes every successful trajectory have negative return. After removing that term and repairing the terminal boundary, one pilot improves by two points. These are development diagnostics, not validated paper contributions.

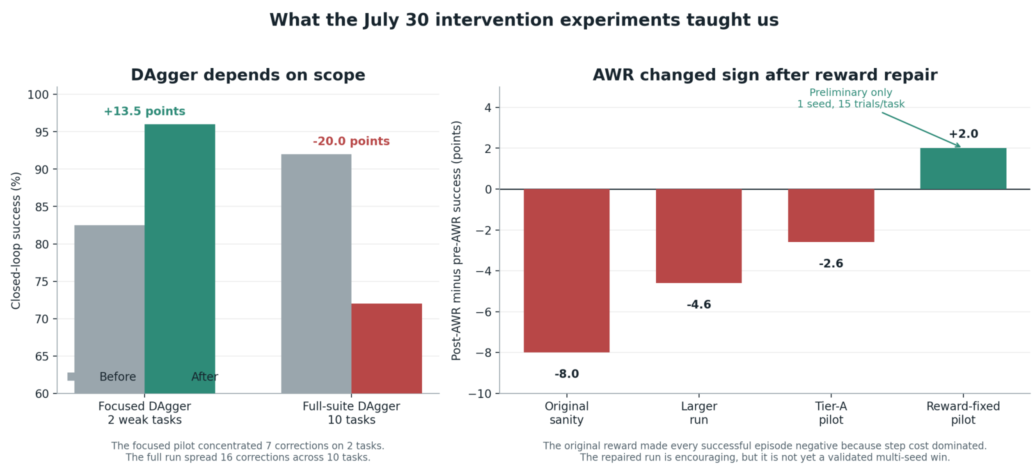

Figure 9: **Development interventions reveal scope and reward sensitivity.** Focused DAgger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

A.7 Expanded attention screen and first direct edit

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-subspace retention preserves $18/20$ successes compared with $16/20$ for both random controls. This 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves $18/20$, compared with $17/20$ for both random controls, which is the wrong direction for the necessity prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and value column edits.

A subsequent discovery sweep evaluates blocks 0, 2, 4, 6, and 7 at ranks 64, 128, and 256. None of the fifteen configurations passes both prespecified 15-point gates. The best configuration, block 0 at rank 128, has a 10-point retention advantage and a 40-point removal advantage over Frobenius-matched random edits. It therefore does not advance to held-out confirmation. The selected keep and removal operations change 56.4% and 82.6% of the affected query, key, and value weights in relative Frobenius norm. This is a global functional ablation, not a small semantic edit.

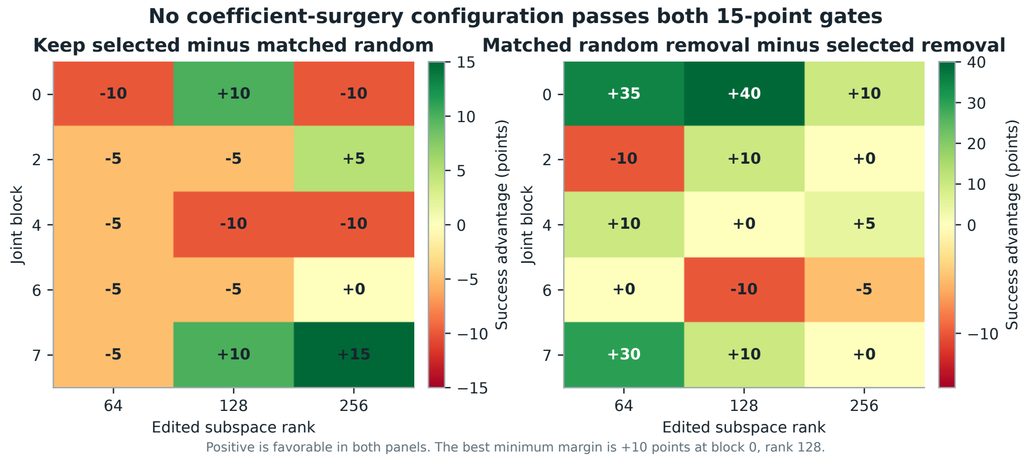

Figure 10: **The complete coefficient-surgery discovery sweep is negative.** Positive values favor the selected weight-derived subspace. No block-rank pair reaches the prespecified 15-point margin in both retention and removal.

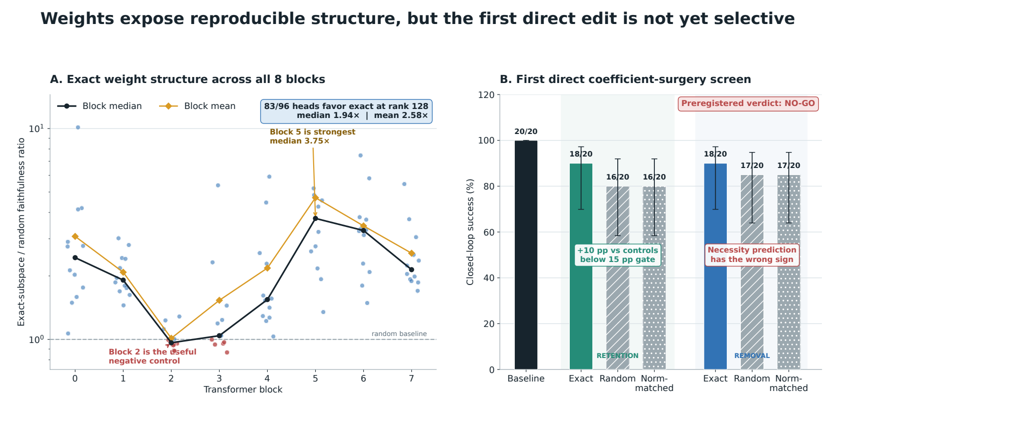

Figure 11: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

A.8 Numerical scope of rational conversion

The deployed Padé computation is exactly rational even though it approximates RMS normalization. Final certification must report its polynomial degrees, coefficients, denominator margin, observed activation range, tail error, action drift, and runtime overhead. Operator membership alone does not establish practical numerical safety under distribution shift.